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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME1241-1- ENERGY MANAGEMENT
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Inexhaustible energy sources are known as**
A) Commercial Energy
B) Renewable Energy
C) Primary energy
D) Secondary energy
2. **World oil reserves are estimated to last over**
A) 47 years
B) 60 years
C) 200 years
D) 75 years
3. **"The judicious and effective use of energy to maximize profits and enhance competitive positions". This can be the definition of:**
A) Energy conservation
B) Energy management
C) Energy policy
D) Energy Audit
4. **The ratio of current year's production to the reference year's production is called as**
A) Demand factor
B) Production factor
C) Utilisation factor
D) Load factor
5. **Which instrument is used to monitor O₂, CO in flue gas? (EA)**
A) Combustion analyzer
B) Power analyzer
C) Pyrometer
D) Fyrite
6. **Lux meter is used to measure.....**
A) Illumination level
B) Sound intensity and illumination level
C) Harmonics
D) Speed
7. **Energy manager should be well versed with**
A) Manufacturing and processing skills
B) Managerial and technical skills
C) Technical and marketing skills
D) Managerial and commercial skills
8. **As per the Energy Conservation Act, 2001, "the verification, monitoring and analysis of use of energy including submission of technical report containing recommendations is called**
A) Energy audit
B) Energy policy
C) Energy conservation
D) Energy efficiency
9. **For a cement plant the parameter, "kWh/MT of clinker "indicates**
A) Energy Index parameter
B) Utility factor
C) Production factor
D) Load factor
10. **Energy management is a key component of**
A) Environmental management
B) Carbon management
C) Nitrogen management
D) Water management
11. **Sankey diagram is an useful tool to represent**
A) Financial strength of the company
B) Management philosophy
C) Input and output energy flow
D) Human resource strength of the company

12. In cogeneration, the system efficiencies can go up to -----
 A) 70% B) 80%
 C) 90% D) 60%
13. In a glass industry, exhaust gas from the glass melting furnace is used for power generation by installing steam boiler and turbine. Then the type of co-generation is called as:
 A) Gas turbine B) Diesel generator
 C) Topping cycle D) Bottoming cycle
14. Recuperator is used mainly as a waste heat recovery system in a ____
 A) Boiler B) Billet Reheating Furnace
 C) Compressor D) Condenser
15. In a low to medium temperature waste heat recovery system which of the device is most suitable
 A) Economizer B) Heat wheels
 C) Air preheater D) Recuperator
16. The heat recovery device in which high conductivity bricks are used for storing heat is
 A) heat pipe B) heat pump
 C) thermo compressor D) regenerator
17. Which of the following is a Greenhouse Gas?
 A) Carbon Monoxide B) Water Vapour
 C) Nitrogen dioxide D) Ethane
18. The ozone depletion process is due to
 A) Carbon dioxide B) Chlorine atoms destroying ozone molecules
 C) UV light D) CHC & PFC molecules
19. Acid rain is caused by the release of the following components from combustion of fuels.
 A) SO_x and NO_x B) SO_x and CO₂
 C) CO₂ and NO_x D) H₂O
20. Both the selling and buying company wins on this flexibility that trade offers with positive effects on economy, resource efficiency and climate is called
 A) Carbon trading B) Carbon fund
 C) Kyoto protocol D) Clean development mechanism

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List the classification of energy resources with examples.	8	L*1	1	1
	b) Define energy audit. Explain different types of energy audit briefly.	8	L2	2	1
2.	a) Explain the simple energy conservation opportunities applicable to domestic sector.	8	L2	1	1
	b) Discuss the functions of energy manager.	8	L2	2	1
3.	a) Explain the Indian energy scenario briefly.	8	L2	1	1
	b) Explain the steps in Energy management.	8	L2	2	1
Unit – II					
4.	a) Explain any four methods of efficient use of steam in boiler distribution system.	8	L2	3	1
	b) Explain the principle of Cogeneration. Mention the merits and demerits of cogeneration.	8	L2	4	1
5.	a) What are the heat losses in boiler? Draw heat balance sheet for the boiler.	8	L1	3	1
	b) Explain Heat pipe waste heat recovery system with neat sketch.	8	L2	4	1
6.	a) Define boiler efficiency. Mention the factors affecting Boiler performance.	8	L2	3	1
	b) What do you understand by the term waste heat? Mention the sources of waste heat and the uses of waste heat recovery.	8	L2	4	1

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | What are the principal air pollutants? Explain briefly the sources and ill effects of air pollutants. | 8 | L2 | 5 | 1 |
| b) | Write a note on Kyoto protocol. | 8 | L2 | 5 | 1 |
| 8. a) | Describe greenhouse effect and also mention the ill effects of global warming. | 8 | L2 | 5 | 1 |
| b) | Explain the following: | | | | |
| | i) Carbon trading | | | | |
| | ii) Energy rating | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
